SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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1. TEAM AND PROJECT OVERVIEW

Team Name

Extragorjoss

Project / Solution Name

CashGuard Risk

Name (Leader)

MUHAMMAD AMEER AMSYAR BIN HAMDAN

Email & Phone (Leader)

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University Affiliation(s)

Universiti Teknologi MARA (UiTM); Universiti Teknikal Malaysia Melaka (UTeM)

Live Prototype URL

https://cash-guard.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/hecctarr/CashGuard

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Flow: invoice arrives (WhatsApp or upload) → fields read and checked → staff confirm → risk engine → owner/manager decides → supplier status update → insert-only audit log.

1) **WhatsApp invoice bot (Flask webhook):** suppliers send an invoice photo, or type INVOICE and fill a form; the bot builds the invoice PDF (ReportLab) with defaults and checks sender, duplicates and bank-detail changes. Phase 1 demo uses the in-app WhatsApp simulator; Twilio/Meta connection is built but not live.

2) **Invoice reading:** Phase 1 uses pre-read fields for demo invoices plus staff confirmation. An OpenAI-compatible vision model (GLM-OCR on RunPod) is coded and planned for the next round.

3) **Risk engine (our Python):** Kaplan-Meier + Bühlmann credibility, 1,000-path Monte Carlo, safest pay date, plain-language explanation. No AI decides.

4) **Streamlit app on Replit:** password sign-in (salted hashes), role-locked Owner/Manager/Employee pages, spending limits, Help on every page.

5) **Data:** CSV/JSON demo store; Supabase schema designed for production; code on GitHub.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.13	Main programming language	PSF License
Streamlit 1.66	Web app: screens, roles, settings, approvals	Apache 2.0
pandas 3.0 · NumPy 2.5 · Plotly 7.1	Data handling; Monte Carlo and survival-analysis maths; charts	BSD-3 · BSD-3 · MIT
Flask 3.1 · Requests 2.34	WhatsApp bot webhook and HTTP calls (live APIs planned)	BSD-3 · Apache 2.0
ReportLab 4 · segno 1.6 · Pillow 12.3	Generated invoice PDFs; supplier share QR code; images	BSD · BSD-3 · MIT-CMU
pytest 8 · Playwright 1.56	15 unit tests; browser testing (development only)	MIT · Apache 2.0

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Meta WhatsApp Cloud API (Meta) — optional	Alternative WhatsApp channel supported in code; not used in the demo	Free tier (not used in demo)
No other external APIs	No payment, maps or public GenAI API is called. No real money is moved	—

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit (Replit)	Hosts the Streamlit app and WhatsApp bot webhook; live prototype URL; keys in Replit Secrets	Sponsor-provided plan
GitHub (GitHub)	Code repository shared with judges	Free
Local file storage	Demo data in CSV/JSON on the app server (reset any time)	Included with host

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Kaplan-Meier survival model (payment delays)	Our own code (engine/payments.py)	Self-trained on synthetic invoices; statistical, not ML	Our own work
Bühlmann credibility weighting	Our own code (engine/payments.py)	Self-calculated from synthetic data	Our own work
Monte Carlo cash simulation (1,000 paths)	Our own code (engine/montecarlo.py)	Simulation using the models above	Our own work

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Demo business data (fictional café)	make_data.py (our repo)	Synthetic	Our own work	N
Customer invoices + payment history (120)	make_data.py	Synthetic	Our own work	N
Daily sales history	make_data.py	Synthetic	Our own work	N
Demo supplier invoices (2 images)	capture/ make_demo_invoices.py	Synthetic (fictional companies)	Our own work	N
Sample bank statement CSV · supplier profiles	demo_invoices/ · make_data.py	Synthetic (fictional bank details)	Our own work	N
Impact-test scenarios (50)	engine/impact.py	Synthetic	Our own work	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude (Anthropic), Claude app / Cowork	Writing and debugging Python code (risk engine, WhatsApp bot, app), tests, README, Supabase schema, demo page, documents	Major

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

A human team member read and ran every piece of AI-assisted code before keeping it. We tested the full app in a browser for all three roles, ran 15 unit tests (including sign-in and page access), checked the WhatsApp invoice form and generated PDFs, and verified the risk engine with a backtest, a calibration table and an out-of-sample impact test (14.4% → 3.3% shortfall rate on synthetic data). All numbers in our pitch come from the running app. The team, not the AI tool, is responsible for the design, code and claims.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Hosting platform	Hosts the live prototype (Streamlit app + webhook)

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Not essential: **Replit** hosts our live Streamlit prototype, but the app runs on any Python host or a laptop. **RunPod is not used in Phase 1**; invoice reading uses pre-read demo fields and staff confirmation, and the RunPod model is planned for the next round. The core risk engine (Kaplan-Meier, Monte Carlo, safest date) runs fully on our own Python code with no sponsor service.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The prototype uses **synthetic, fictional data only**. Users sign in with a username and password; passwords are stored only as salted PBKDF2 hashes, and each role can open only its own pages. Demo data is stored in CSV/JSON files on the app server and can be reset.

In real use the app stores business finance data (bills, customer invoices, cash), supplier WhatsApp numbers and bank details. Planned protection: database Row Level Security (Supabase), private storage for invoice files and webhook signature checks. Already built: insert-only audit log, bank-change warnings and supplier updates that never show cash figures. Keys live in Replit Secrets, never in the repository. We follow PDPA principles.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Reading errors: invoice fields can be wrong. Mitigation: confidence flags, line items must add up, duplicate checks, and staff confirm every field.

Form typos: suppliers may type items wrongly. Mitigation: the bot rejects unreadable lines and asks them to fix and resend; staff still confirm.

Invoice fraud: changed bank details are flagged for a phone check before paying.

Not yet live: AI invoice reading (RunPod) and live WhatsApp are planned; Phase 1 uses demo fields and the in-app simulator.

Little history / synthetic data: credibility blending and a rule-based timeline; results need real-world validation.

Scope: decision support only. No automatic payments; AI never approves anything.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]

25 October 2026